

Practical 1: The Journey of a Photon Through the Expanding Universe

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Objectives

In this practical session we will explore the relation between the expansion history of the Universe, cosmological distances, cosmic time, and the evolution of the different energy-density components.

Rather than treating these quantities as independent formulae, we will follow the journey of a photon emitted by a distant galaxy and received by an observer today.

By the end of the practical you should be able to:

- construct the expansion history $H(z)$ for a given cosmology;
- calculate cosmological distances using numerical integration;
- relate redshift to cosmic and lookback time;
- understand the physical meaning of different cosmological distances;
- determine how matter, radiation, and dark-energy densities evolve;
- identify the main epochs in the expansion history of the Universe;
- make qualitative predictions before performing numerical calculations;
- test whether a numerical result is physically reasonable.

General instructions

The purpose of this practical is not simply to obtain the correct numerical values or plots. You should be able to explain the physical and numerical reasoning behind your results.

For several parts of the practical, you will be asked to make a prediction **before running your code**. Record your prediction before performing the calculation.

The use of AI tools is not forbidden; however, you are responsible for understanding every part of the work you submit. During the practical session, the instructor may ask you to explain or modify your calculation, or to predict the effect of a modification before running the code.

Unless otherwise stated, do not use CAMB, CLASS, Astropy Cosmology, or another package that directly calculates the cosmological quantities requested below.

1 The cosmological model

Consider initially a spatially flat Universe containing matter and a cosmological constant,

$$\Omega_{m,0} = 0.3, \quad \Omega_{\Lambda,0} = 0.7. \quad (1)$$

Assume

$$H_0 = 100 h \text{ km s}^{-1} \text{ Mpc}^{-1}, h = 0.7. \quad (2)$$

Before coding

Write down the expression for

$$H(z) \quad (3)$$

for this cosmological model.

Answer the following questions without performing a numerical calculation:

1. Was the expansion rate $H(z)$ larger or smaller in the past?
2. Which component dominates $H(z)$ at sufficiently large redshift?
3. Which component will dominate the expansion at sufficiently late times?

2 A photon emitted at high redshift

Consider a photon emitted by a galaxy at

$$z_{\text{em}} = 3 \quad (4)$$

and observed by us today at $z = 0$.

Our goal is to reconstruct several aspects of the photon's journey.

2.1 When was the photon emitted?

The lookback time is the time elapsed between the emission of the light we observe from an object and its reception by us today.

The relation between cosmic time and redshift can be written as

$$dt = -\frac{dz}{(1+z)H(z)}. \quad (5)$$

Therefore, the lookback time to an object observed at redshift z is

$$t_{\text{LB}}(z) = \int_0^z \frac{dz'}{(1+z')H(z')}. \quad (6)$$

Prediction

Before performing the calculation, answer:

1. Is the travel time of the photon simply equal to the comoving distance to the galaxy divided by c ?
2. Explain your answer physically.

Numerical calculation

Calculate the lookback time for the photon emitted at $z_{\text{em}} = 3$.

Express the result in Gyr.

Also calculate $t_{\text{LB}}(z)$ over the interval

$$0 \leq z \leq 3 \tag{7}$$

and plot it as a function of redshift.

2.2 How far has the photon travelled?

The line-of-sight comoving distance is

$$D_C(z) = c \int_0^z \frac{dz'}{H(z')}. \tag{8}$$

Calculate $D_C(z)$ from $z = 0$ to $z = 3$ and express the result in Gpc/h.

Report the value at $z = 3$.

Compare the following two quantities:

$$D_C(z_{\text{em}}), \quad c t_{\text{LB}}(z_{\text{em}}). \tag{9}$$

Are they equal?

Explain physically why or why not.

3 Three different cosmological distances

For a spatially flat Universe, calculate

$$D_C(z), \tag{10}$$

$$D_A(z) = \frac{D_C(z)}{1+z}, \tag{11}$$

and

$$D_L(z) = (1+z)D_C(z). \quad (12)$$

Calculate the three distances over

$$0 \leq z \leq 3 \quad (13)$$

and plot them in the same figure in units of Gpc/h.

Prediction before plotting

Before producing the figure:

1. Which of the three distances do you expect to increase monotonically?
2. Do you expect any of them to reach a maximum?
3. Sketch qualitatively the expected behavior of the three curves.

After producing your numerical result, compare it with your prediction.

If one of the distances has a maximum, determine approximately the redshift at which the maximum occurs.

3.1 Why do we need different distances?

Consider identical galaxies with a physical transverse size

$$L = 10 \text{ kpc}. \quad (14)$$

Their observed angular size is

$$\theta(z) = \frac{L}{D_A(z)}. \quad (15)$$

Calculate and plot $\theta(z)$ over the range

$$0.1 \leq z \leq 3. \quad (16)$$

Before plotting the result, answer:

If identical galaxies are placed progressively farther away from us, should their apparent angular size always decrease?

Use your numerical result to determine approximately the redshift at which a 10 kpc galaxy has its smallest apparent angular size.

Explain the result physically.

3.2 A luminosity-distance observer

Suppose instead that an astronomer knows the intrinsic luminosity L_{gal} of the galaxy and measures its flux F .

The luminosity distance is defined through

$$F = \frac{L_{\text{gal}}}{4\pi D_L^2}. \quad (17)$$

Explain why an astronomer using angular sizes and another astronomer using fluxes assign different cosmological distances to the same galaxy.

Verify numerically the relation between D_L and D_A for your cosmological model.

4 What happened to the Universe while the photon was travelling?

Matter evolves as

$$\rho_m(a) = \rho_{m,0} a^{-3}, \quad (18)$$

while the density associated with the cosmological constant remains constant,

$$\rho_\Lambda(a) = \rho_{\Lambda,0}. \quad (19)$$

Determine analytically the scale factor and redshift at which

$$\rho_m = \rho_\Lambda. \quad (20)$$

Denote them by

$$a_{\text{eq}}^{m,\Lambda}, \quad z_{\text{eq}}^{m,\Lambda}. \quad (21)$$

Then determine the corresponding lookback time numerically.

Answer:

1. Was the Universe matter dominated when the photon at $z = 3$ was emitted?
2. Did matter- Λ equality occur while the photon was travelling?
3. Does equality between two densities represent an instantaneous transition from one cosmological era to another? Explain.

5 A photon from the early Universe

We now extend the calculation to much earlier times.

Include radiation and consider

$$\Omega_{m,0} = 0.2999, \quad \Omega_{\Lambda,0} = 0.7, \quad \Omega_{R,0} = 10^{-4}. \quad (22)$$

The expansion rate is now

$$H(z) = H_0 \sqrt{\Omega_{R,0}(1+z)^4 + \Omega_{m,0}(1+z)^3 + \Omega_{\Lambda,0}}. \quad (23)$$

Imagine a photon emitted at

$$z_{\text{em}} = 10^4. \quad (24)$$

5.1 Cosmological archaeology

The different components evolve according to

$$\rho_R(a) \propto a^{-4}, \quad (25)$$

$$\rho_m(a) \propto a^{-3}, \quad (26)$$

$$\rho_\Lambda(a) \propto \text{constant}. \quad (27)$$

Determine analytically:

$$a_{\text{eq}}^{R,m}, \quad z_{\text{eq}}^{R,m}, \quad (28)$$

and

$$a_{\text{eq}}^{m,\Lambda}, \quad z_{\text{eq}}^{m,\Lambda}. \quad (29)$$

Use your numerical calculation of $t(z)$ to determine the corresponding cosmic or lookback times.

Answer:

Through which cosmological regimes has a photon emitted at $z = 10^4$ travelled before reaching us today?

6 Evolution of matter density with cosmic time

For each limiting regime, determine analytically the expected dependence of the scale factor on cosmic time,

$$a(t), \quad (30)$$

and use

$$\rho_m \propto a^{-3} \quad (31)$$

to derive the expected evolution of

$$\rho_m(t). \tag{32}$$

Prediction

Before making the numerical plot:

1. Determine the expected slope of $\rho_m(t)$ in log–log space during radiation domination.
2. Determine the expected slope during matter domination.
3. What behavior do you expect once the cosmological constant dominates?

Numerical calculation

Construct numerically the evolution of

$$\rho_m(t) \tag{33}$$

between $z = 10^4$ and $z = 0$.

Plot $\rho_m(t)$ versus cosmic time using logarithmic axes.

On the same figure:

- indicate radiation–matter equality;
- indicate matter– Λ equality;
- indicate the analytical asymptotic behaviors predicted above.

Discuss where the numerical solution follows the limiting analytical solutions and where it departs from them.

7 Numerical and physical checks

A numerical result should not be accepted simply because the code runs and produces a smooth curve.

Design at least **two independent checks** of your calculations.

At least one should be a physical or analytical check rather than simply repeating the calculation using another numerical routine.

Explain:

1. what you are checking;
2. what result you expect;
3. what would indicate that something is wrong with your calculation.

8 Individual challenge

During the presentation session, the instructor can give you an individual modification or question related to your calculation.

You will first be asked to predict the effect of the modification **without running your code**.

You will then modify or use your calculation to test your prediction.

Possible modifications may involve:

- changing the redshift of the source;
- changing one cosmological parameter;
- identifying an error in a calculation;
- comparing two cosmological epochs;
- predicting how an observable will change;
- diagnosing an apparently reasonable but incorrect numerical result.

The objective is not to obtain an immediate correct answer, but to demonstrate that you can reason about the physical model and use your numerical calculation to test that reasoning.

AI-use statement

At the end of your work, briefly state:

- whether you used an AI tool during the practical;
- what you used it for;
- one suggestion or result from the AI that you independently checked;
- how you verified it.

What should you submit?

Submit your notebook or code together with a short report containing:

- your predictions made before the numerical calculations;
- the requested plots and numerical results;
- your physical interpretation of the results;
- your analytical calculations;
- your numerical and physical consistency checks;
- your AI-use statement.

The quality of the practical will be evaluated not only from the final plots, but also from your ability to explain why the results have the behavior you observe.